# The Beer–Lambert Law

Pupil sheet

**Name:** _____ **Date:** _____

**Safety:** The box is not waterproof: keep water away from it. Goggles with dyes; taste nothing.

## 1. What I see on the screen

The screen shows **Illuminance I** (in lux) and **Absorbance A**; I choose the sample in the **Sample in the beam:** list and press **Capture**. **Only the illuminance is measured**; the absorbance is computed by the app.

**Careful — the reference.** I put the **water blank** in the beam and press **Set reference $I_0$ from this reading**. Below the cards I check that $I_0$ was *chosen by me*; if not, I take it again.

**That is why I write the lux in the table, not the absorbance from the screen.** From the lux I can rebuild everything; from the screen's absorbance I cannot.

$I_0$ **(the blank) =** _____ **lx**
Write it down *before* moving on: it does not appear in the exported file.

## 2. What I do on my own

**First I predict, before any measuring.** I draw two sketches. I add the real curve over them, in another colour, at the end of the lesson.

Transmitted light ($I$) against $c$

$A = \log_{10}(I_0/I)$ against $c$

**The rig** (measured once, then nothing moves): lamp-to-sensor distance _____ cm path length $l$ of the cup _____ cm

**I measure.** The blank first. The volumes are already written in. **The first row is worked as an example** — I use it as the pattern for the rest.

| $c$ | **Stock** | **Water** | $I$ **(lx)** | $T = I/I_0$ | $A = -\log_{10} T$ |
|---|---|---|---|---|---|
| *example* | | | *1000* | *0.50* | *0.301* |
| 0 (blank) | 0 mL | 100 mL | | 1.00 | 0.00 |
| 0.2 | 20 mL | 80 mL | | | |
| 0.4 | 40 mL | 60 mL | | | |
| 0.6 | 60 mL | 40 mL | | | |
| 0.8 | 80 mL | 20 mL | | | |
| 1.0 | 100 mL | 0 mL | | | |
| $\approx 2$ | — | — | | | |
| 0 (end) | 0 mL | 100 mL | | | |

*Check: $T = 10\%$ must give $A = 1$. If I get a negative number, I used percentages instead of a fraction.*

**The line.** The gradient, computed on the linear points only: _____. The points leave the line from about $A \approx$ _____.

**How sure are we?** The app's slope: _____ ± _____, $R^2$ = _____. The blank at the end differs from $I_0$ by _____ %. Is the series good? ☐ yes ☐ no

## 3. The conclusion I write

I predicted that, and the data show that _____

_____

The logarithm straightens the graph because _____

_____

Our instrument is trustworthy up to $A$ = _____, and above that, because _____

_____

**Exit ticket**

Why does the logarithm straighten the graph?

_____